GHENT UNIVERSITY
MODELLING AND SIMULATING OF DYNAMIC SYSTEMS
SPRING 2026

Co-design of biped walkers

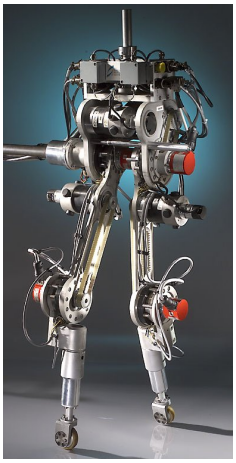

Figure 1: *Photograph of the RABBIT biped walker system. Image courtesy of Jessy Grizzle.*

1 Problem Statement

Biped walkers are systems that depend on legged locomotion to navigate their environment. These systems have been studied extensively in the last couple of decades. The main advantage of legged over wheeled locomotion is the dynamic reprogrammability and resulting multifunctionality of limbs and limb based locomotion. By adapting the walking gait (dynamically), a variety of terrains and obstacles can be navigated.

Although the design of biped walkers is obviously inspired by human example, the morphology of biped walkers can differ substantially from their biological predecessor. Take as an example the iconic RABBIT biped walker depicted in Fig. 1. The ratio between tibia (the lower leg) and femur (the upper leg) equals 1 whereas the ratio in humans equal 0.8 on average.

The human body co-evolved with the ways we adopted to navigate our environment. This means that our bodily design is adjusted so as to optimally support the variety of dynamic behaviours we (used to) exhibit in day to day live. In engineering literature, concurrent optimization of a system's morphology and control system is referred to as co-design.

The biped literature usually focusses on designing stable walking gaits by a combination of motion planning and advanced feedback control design for a fixed morphology of the system. In this project we will design an energy efficient gait and optimal tibia/femur ratio concurrently.

2 Modelling

For this study on biped walkers we will restrict ourselves to a rudimentary model existing of 5 links. Two links are used to model each leg, the fifth link is used to model the torso. We do not consider arms nor feet. The feet are approximated as passive points contacts that do not slip when in contact with the ground.

The model is planar, left-right symmetric and tailored to walking gaits. Walking gaits are characterised by the condition that at least one foot has contact with the ground. Actuation is preserved for the two knee joints and the two hip joints. This means the system is effectively under-actuated.

A gait can be decomposed in two symmetric phases where either the left or right leg has contact with the ground surface. The transition between these two phases is instantaneous and occurs when the free leg makes contact with the ground surface. After the instantaneous moment where both feet are in contact with the support surface it is assumed that the supporting leg and free leg have switched place meaning now the supporting leg is free and the free leg is supporting.

This section covers detail regarding the kinematic and dynamic model the 5 link biped model. The qualitative behaviour of dynamics is modelled using a continuous single-stance dynamic model and an instantaneous or discrete impact heel-strike impact model.

2.1 Kinematics

The kinematics of the system in single-stance configuration are presented schematically in Fig. 2.

We use 5 generalised coordinates q_1, q_2, q_3, q_4 and q_5. These 5 coordinates are combined in a vector $\mathbf{q}$. The angle q_5 is absolute and denotes the torso's rotation with respect to the inertial frame. Angles q_1 and q_2 represent the relative pose of the supporting and free femur with respect to the torso, angles q_3 and q_4 represent the relative pose of the supporting and free tibia.

Points corresponding with the lower end of the links are labelled with P_n. Note that P_3 is stationary by definition and can be considered as the origin. Points corresponding with the centre of mass of each link are denotes with G_n.

Starting from P_3 the position of the different points of interest can be represented as a function of the generalised coordinates. These functions represent the kinematics of the system and can be worked out manually.

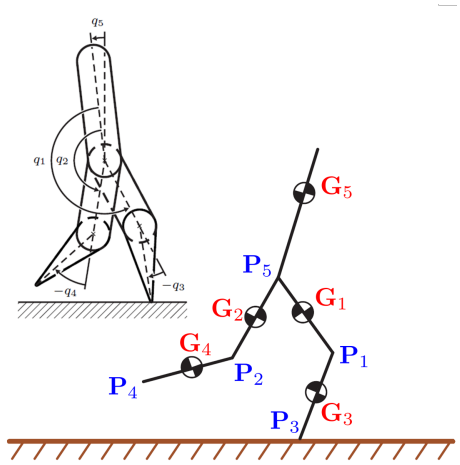

Figure 2: *Schematic representation of a 5 link biped model. Labelling of points on the robot is so that labels* 1 *and* 3 *always refer to the supporting tibia and femur,* 2 *and* 4 *refer to the free tibia and femur and* 5 *refers to the torso.*

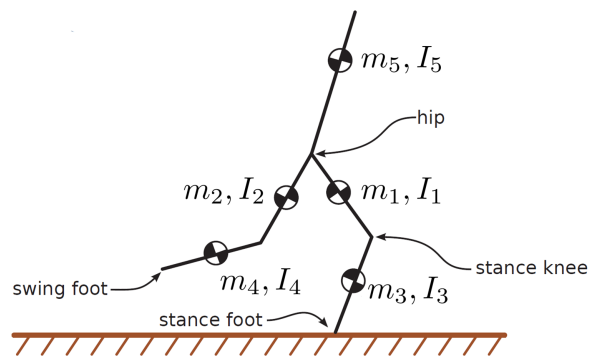

Figure 3: *Schematic representation of single-stance continuous dynamics.*

2.2 Single-stance continuous dynamics

The single stance continuous dynamics can be derived using the Euler-Lagrangian equations. Note here that the angle q_5 is not controlled, only the relative angles q_1, q_2, q_3 and q_4 are controlled using motor torques u_1, u_2, u_3 and u_4. These input variables are stacked in the vector $\mathbf{u}$.

The total kinetic energy of the systems is determined by the translational or linear kinetic energy and rotational or angular kinetic energy.

$$T = T_l + T_r = \tfrac{1}{2}\sum_{n=1}^{5} m_n \|\dot{G}_n\|^2 + \tfrac{1}{2}\sum_{n=1}^{5} I_n \dot{\theta}_n \tag{1}$$

Here θ_n represents the absolute angle of each link. The velocities $\dot{G}_n$ can be derived by differentiating the kinematic functions.

The potential energy is given by

$$V = \sum_{n=1}^{5} m_n g \mathbf{e}_y^\top G_n \tag{2}$$

with g the gravitational constant.

Applying the Euler-Lagrange formalism produces a coupled system of second order differential equations of the following form with generalised velocities, $\dot{\mathbf{q}}$, and accelerations, $\ddot{\mathbf{q}}$. This system of equations governs the single-stance continuous dynamics of the 5 link biped system.

$$\mathrm{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{c}(\mathbf{q}, \dot{\mathbf{q}}) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \mathbf{u} = \mathrm{B}\mathbf{u} \tag{3}$$

In robotics literature, the nonlinear coupling matrix function, M, is referred to as the mass matrix. The vector function, $\mathbf{c}$, combines the effects of gravitation and of centrifugal and coriolis effects.

Considering the following state definition

$$\mathbf{x} = \begin{bmatrix} \mathbf{q} \\ \dot{\mathbf{q}} \end{bmatrix} \tag{4}$$

the system of equations can be moulded into state-space form

$$\dot{\mathbf{x}} = \mathbf{f}_{\mathrm{SS}}(\mathbf{x}, \mathbf{u}) = \begin{bmatrix} \dot{\mathbf{q}} \\ \mathrm{M}(\mathbf{q})^{-1}(\mathrm{B}\mathbf{u} - \mathbf{c}(\mathbf{q}, \dot{\mathbf{q}})) \end{bmatrix} \tag{5}$$

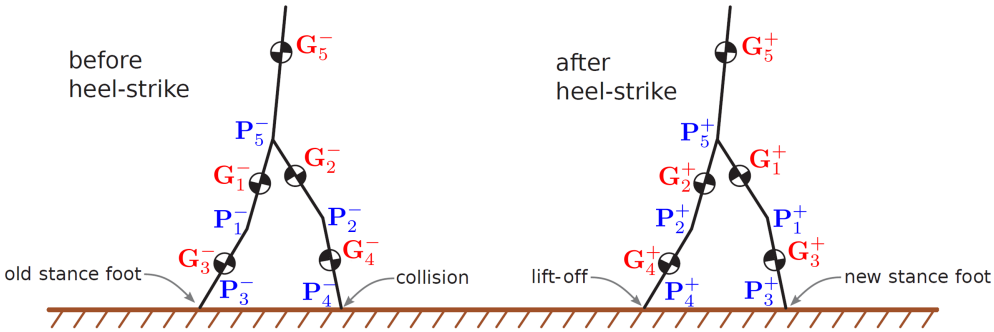

Figure 4: *Schematic representation of heel-strike impact model before $-$ and after $+$ impact. The points on the robot are relabelled after impact, reflecting the left-right symmetry of the robot.*

2.3 Heel-strike impact model

The heel-strike impact model governs the relation between the generalised coordinates and velocities before and after impact. Recall that we assumed the walker switches between its supporting and free leg instantaneously. We further assume that the impact force has no result on the robot's configuration.

Assume, without loss of generality, the impact takes place at time 0. Then we can formally define the state before, $\mathbf{x}^-$, and after impact, $\mathbf{x}^+$, through the following limits.

$$\begin{aligned} \mathbf{x}^- &= \lim_{t \uparrow 0} \mathbf{x}(t) \\ \mathbf{x}^+ &= \lim_{t \downarrow 0} \mathbf{x}(t) \end{aligned} \tag{6}$$

Now the impact model can be formalized as

$$\mathbf{x}^+ = \mathcal{I}(\mathbf{x}^-) \tag{7}$$

The configuration of the system is equal before and after impact though the supporting and free leg switch place. In order to use the same single-stance model from the previous section to describe the governing dynamics after impact, the following transformation is applied to the generalised coordinates before impact, $\mathbf{q}^-$, to yield the generalised coordinates after impact, $\mathbf{q}^+$. Due to the left-right symmetry of the system, the configuration remains the same however the supporting and free leg will indeed have switches place.

$$\mathbf{q}^+ = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \mathbf{q}^- \tag{8}$$

In order to determine the corresponding generalised velocities before, $\dot{\mathbf{q}}^-$, and after, $\dot{\mathbf{q}}^+$, impact, consider the following relation. This relation expresses that the generalized momentum before and after the collision is constant.

$$\mathrm{M}(\mathbf{q}^+)\dot{\mathbf{q}}^+ = \mathrm{M}(\mathbf{q}^-)\dot{\mathbf{q}}^- \tag{9}$$

Note that this implies the impact model is inelastic and results into a sudden decrease of kinetic energy. Although this is not immediately clear from this equation, note that the mass matrix implicitly encodes which is the supporting leg and resulting which feet is stationary.

symbol	value	name
m_3, m_4	$3.2\,\mathrm{kg}$	mass of tibia (lower leg)
m_1, m_2	$6.8\,\mathrm{kg}$	mass of femur (upper leg)
m_5	$20\,\mathrm{kg}$	mass of torso
I_3, I_4	$0.93\,\mathrm{kg\,m^2}$	rotational inertia of tibia, about its centre of mass
I_1, I_2	$1.08\,\mathrm{kg\,m^2}$	rotational inertia of femur, about its centre of mass
I_5	$2.22\,\mathrm{kg\,m^2}$	rotational inertia of torso, about its centre of mass
l_3, l_4	$0.4\,\mathrm{m}$	length of tibia
l_1, l_2	$0.4\,\mathrm{m}$	length of femur
l_5	$0.625\,\mathrm{m}$	length of torso
r_3, r_4	$0.128\,\mathrm{m}$	distance from tibia centre of mass to knee
r_1, r_2	$0.163\,\mathrm{m}$	distance from femur centre of mass to hip
r_5	$0.2\,\mathrm{m}$	distance from torso centre of mass to hip

Table 1: *Lumped parameter values of the 5 link biped walker model.*

Task 1: Modelling, simulation & visualization

Implement the kinematic, single-stance continuous dynamics and heel-strike impact models of the 5 link biped walker in computational environment of your choice (e.g. Matlab, Python, Julia, etc.). The lumped model parameters are given in Table 1.

To determine the Euler-Lagrange model we strongly recommend the use of a symbolic package (e.g. Symbolic Math Toolbox in Matlab, CasADi in Python, etc.). Make sure that you can evaluate the mass matrix function separately provided that it is used in both the single-stance continuous dynamics formalism as well as the heel-strike impact model.

Work towards the following two deliverables

- Implement a visualization routine that enables you to visualize the configuration of the biped walker. Additionally it is recommended that you foresee a means of visualizing the biped walker dynamically. This routine is valuable to verify visually whether your model is correct.
- Validate your model by performing a forward simulation using an Ordinary Differential Equation (ODE) solver that is preprogrammed in the computational environment of your choice. Perform the following experiments for validation
  - Gravitational collapse (random initial generalized coordinates, zero initial generalized velocities, zero torques). Do not model the impact.
  - Frozen body (arbitrary initial configuration with the free leg extended in the front, negative generalized velocity $\dot{q}_5$ [rotating clockwise], assume that system is controlled so that only $\ddot{q}_5 \neq 0$, this can be achieved by considering only the last equation from (3) setting $\dot{q}_n \equiv 0$ and $\ddot{q}_n \equiv 0$ for $n \in \{1, 2, 3, 4\}$). Run until impact and continue for $0.5\,\mathrm{s}$ to validate the impact model.

3 Optimization

In this section we describe how to gradually set-up the co-optimization problem. The goal of the co-optimization is to find an optimal walking gait for given step length and step time (fixing the average speed). Additionally we want to find the optimal tibia/femur ratio so that the leg length remains constant.

3.1 Motion optimization

The first step in solving the co-optimization problem is to reformulate the gait optimization problem as an optimal control problem with initial and final state constraint. Such a problem has the following form

$$\begin{aligned}
&\min_{\mathbf{x}(t),\mathbf{u}(t)} \int_0^T l(\mathbf{x}(t),\mathbf{u}(t))\mathrm{d}t \\
&\text{s.t. } \dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t),\mathbf{u}(t)) \\
&\qquad 0 = \mathbf{g}_0(\mathbf{x}(0)) \\
&\qquad 0 = \mathbf{g}_T(\mathbf{x}(T)) \\
&\qquad 0 = \mathbf{g}_{0\to T}(\mathbf{x}(0),\mathbf{x}(T)) \\
&\qquad 0 \leq \mathbf{h}_0(\mathbf{x}(0)) \\
&\qquad 0 \leq \mathbf{h}_T(\mathbf{x}(T))
\end{aligned} \tag{10}$$

We can limit the gait optimization to perform a single step. Due to symmetry the next step needs not be considered separately. To find a natural gait with step length L (stride length $2L$) and step duration T consider the following definitions

- The cost rate function is given by
$$l(\mathbf{x},\mathbf{u}) = \tfrac{1}{2}\|\mathbf{u}\|^2 \tag{11}$$
- The dynamics are governed by the single-stance dynamics
$$\mathbf{f}(\mathbf{x},\mathbf{u}) = \mathbf{f}_{\text{SS}}(\mathbf{x},\mathbf{u}) \tag{12}$$
- The initial state is constraint so that at time 0 the free foot is at minus the step length L and has positive vertical velocity.
$$P_5(\mathbf{x}(0)) = \begin{bmatrix}-L & 0\end{bmatrix}^\top \tag{13}$$
$$\mathbf{e}_y^\top \dot{P}_5(\mathbf{x}(0)) > 0 \tag{14}$$
- The final state is constraint so that at time T the free foot has travelled a stride length $2L$ and has negative vertical velocity
$$P_5(\mathbf{x}(T)) = \begin{bmatrix}+L & 0\end{bmatrix}^\top \tag{15}$$
$$\mathbf{e}_y^\top \dot{P}_5(\mathbf{x}(T)) < 0 \tag{16}$$
- To obtain a left-right symmetric periodic gate we introduce a final constraint that relates the initial and final state through the impact map. By definition, the initial state corresponds with the system just after an impact has occurred whereas the final state corresponds with the system just before impact. Hence we have that
$$\mathbf{x}(0) = \boldsymbol{\mathcal{I}}(\mathbf{x}(T)) \tag{17}$$

Task 2: Gait optimization

Use the models from the first task to set-up the optimization problem described above. Solve the problem using the direct multiple shooting code from the exercise session. You can personally pick the step length L and step duration T to introduce some variability amongst the solution of different groups.

3.2 Co-optimization

The gait optimization in the previous section assumed all model parameters were fixed. Formally the gait optimization problem is easily extended to include some parameter vector $\boldsymbol{\theta}$ in the optimization variables. Solution of this problem then produces an optimal gait and associated optimal set of parameters. Note that this presumes sufficient flexibility in the system design to ensure that a design is possible corresponding with these parameters.

There are generally two approaches to optimize the co-design problem.

- In a *simultaneous* approach, the direct multiple shooting formalism is extended with the model parameters. Note that the sparsity of the corresponding nonlinear program (after transcription of the continuous problem) will be hampered resulting in increased computation time.

$$\min_{\boldsymbol{\theta},\mathbf{x}(t),\mathbf{u}(t)} \int_0^T l(\mathbf{x}(t),\mathbf{u}(t))\mathrm{d}t \text{ s.t. ...} \tag{18}$$

- In a *nested* approach, the gait optimization problem is solved for fixed parameters. Note that the resulting cost value is an implicit function of $\boldsymbol{\theta}$. The function can be evaluated only numerically.

$$J(\boldsymbol{\theta}) = \min_{\mathbf{x}(t),\mathbf{u}(t)} \int_0^T l(\mathbf{x}(t),\mathbf{u}(t))\mathrm{d}t \text{ s.t. ...} \tag{19}$$

  A second optimization routine is then used to optimize the gait cost. Note that the optimization procedure need be adapted to the properties of the objective, J.

$$\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) \tag{20}$$

Task 3: Co-optimization task

Determine the optimal tibia/femur ratio with constant leg length and for given step length L and step duration T chosen in taks 2. You can either use a simultaneous approach or a nested approach. In case of the simultaneous approach you will need to adapt the code from the exercise session. In case of the nested approach you can use an available black-box optimization routine.

If the result of the co-design is disappointing, try changing the cost function, e.g. by adding more weight to the knee joints than the hip joints or vice versa.

4 Organisation

We expect you to form teams of four students. Questions and feedback during the semester can be obtained by making an appointment via email with the project supervisor. The project supervisor is Jeroen Taets, Jeroen.Taets@UGent.be.

5 Reporting

The project consists of three large deliverables: two reports and a final pitch. It is mandatory to use either the Word or LaTeX IEEE two column template for your reports. The templates can be found on Ufora. Note that also the introduction to Simulink makes use of this template.

- The **milestone report** should be written with length between 2-4 pages long and should cover at least the modelling part of your project as well as some results of the simulation verifying the model. The milestone report is due on Sunday **19/4**.
- The **final report** should be 4-8 pages long and contain a summary of the modelling and simulation part next to the secondary tasks. A short description of the optimization algorithm that is used and a motivation for choosing this algorithm is required, as well as a result from your simulation verifying the performance of your optimization. The final report is due on Sunday **17/5**.
- The goal of the last pitch is to present and motivate your solution to your peers and the project supervisors. The scope of the final pitch should be less technical than the final report, focussing on a short description of the objective, a brief high-level overview of the methods used and an analysis and discussion of the final results. There is a 5 minute timelimit on the final pitch per group. The pitches are organised in the week after the deadline. Details about the venue and time will be communicated later on.

You can submit your reports via mail.

Plagiarism in both the project code and reports will be checked and punished according to the faculty rules.

6 Timeline overview

The following deadlines are important to keep in mind for the project

- Milestone report is due on **19/04**
- Final report is due on **17/05**
- Final project pitch is organised in Week 13.